IE6600 — Computation and Visualization for Analytics

Project 2: Advanced Data Analysis and Visualization Using Seaborn

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NYC Motor Vehicle Collisions – Crashes

Group	Group 4
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Dataset	Motor Vehicle Collisions – Crashes (NYC Open Data / data.gov)
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1. Project Overview and Objective

This report presents a comprehensive exploratory data analysis (EDA) and statistical investigation of the NYC Motor Vehicle Collisions – Crashes dataset. The project demonstrates proficiency in data acquisition, cleaning, visualization using the Seaborn library, and statistical inference.

Motor vehicle collisions represent a critical public safety concern in New York City. By systematically analyzing collision records, this project aims to uncover temporal patterns, borough-level differences, injury distributions, and the leading contributing factors behind crashes — information that can inform traffic safety policy and urban mobility planning.

Project Covered

- Task 1: Dataset Selection and Confirmation
- Task 2: Data Acquisition and Inspection
- Task 3: Data Cleaning and Preparation
- Task 4: Exploratory Data Analysis (EDA) using Seaborn
- Task 5: Statistical Analysis and Reporting Insights
- Task 6: Advanced Analysis (Optional)

2. Dataset Selection and Confirmation

The dataset selected for this project is the **NYC Motor Vehicle Collisions – Crashes** dataset, sourced from **NYC Open Data** (data.cityofnewyork.us) and indexed on **data.gov**.

Dataset Name	Motor Vehicle Collisions – Crashes
Source	NYC Open Data (data.cityofnewyork.us) / data.gov
Resource ID	h9gi-nx95
Domain	Transportation / Public Safety
Update Freq.	Weekly (live API endpoint)
Access Method	Official CSV API endpoint with selected column projection

The dataset is particularly suitable for this project because it contains rich temporal attributes (crash date and time), spatial attributes (borough, latitude/longitude), injury and fatality counts disaggregated by road user type, and categorical fields such as contributing factors and vehicle types. This variety supports both distributional and relational analysis using Seaborn.

3. Data Acquisition and Inspection

Data was loaded using a parameterized API call to the official NYC Open Data endpoint, retrieving 23 columns relevant to the analysis. When a local CSV file was already available in the working directory, that file was used for reproducibility; otherwise, the live API was queried.

Columns Retrieved

The following column categories were selected from the full dataset to keep the analysis focused and efficient:

- Temporal fields: **crash_date**, **crash_time**
- Geographic fields: **borough**, **zip_code**, **latitude**, **longitude**, street names
- Injury/fatality counts: **persons, pedestrians, cyclists, motorists** (injured and killed)
- Causal fields: **contributing_factor_vehicle_1**, **contributing_factor_vehicle_2**
- Vehicle fields: **vehicle_type_code1**, **vehicle_type_code2**
- Identifier: **collision_id**

Preliminary Inspection Summary

After loading the data, an initial inspection was conducted to understand the structure and content of the dataset. Key observations included:

- The dataset contains hundreds of thousands of collision records spanning multiple years.
- Missing values were concentrated in geographic (latitude/longitude, borough) and causal (contributing factor, vehicle type) columns.
- No exact duplicate records were found after applying the deduplication step.
- Several columns stored as strings required conversion to numeric or datetime formats.

4. Data Cleaning and Preparation

A structured, multi-step cleaning pipeline was applied to ensure the dataset was suitable for reliable statistical analysis and visualization.

4.1 Column Standardization

All column names were normalized by stripping whitespace, converting to lowercase, and replacing spaces with underscores. This prevents ambiguity in column references throughout the analysis pipeline.

4.2 Duplicate Removal

Exact duplicate rows (all columns identical) were identified and removed using pandas *drop_duplicates()*. The dataset was confirmed to contain no meaningful duplicate records.

4.3 Type Conversion

Numeric columns — including **zip_code**, **latitude**, **longitude**, all injury/fatality counts, and **collision_id** — were coerced to numeric format, with non-parseable values replaced by NaN. The **crash_date** column was parsed as a datetime object, and **crash_time** was converted to a time object using the %%H:%%M format.

4.4 Categorical Cleaning

Categorical columns (**borough**, street names, contributing factors, vehicle types) were stripped of leading/trailing whitespace. Placeholder values — '', 'nan', empty strings, and 'Unspecified' — were replaced with proper NaN values. Borough names were title-cased for consistent display in visualizations.

4.5 Feature Engineering

The following derived features were created to support the EDA:

Derived Feature	Description	Type
crash_hour	Hour of day extracted from crash_time	Integer (0–23)
year	Year extracted from crash_date	Integer
month	Month number extracted from crash_date	Integer (1–12)
month_name	Full month name	String
day_of_week	Day name (Monday–Sunday)	String
is_weekend	True if Saturday or Sunday	Boolean
total_injured	Persons injured per collision	Float
total_killed	Persons killed per collision	Float
total_casualties	Sum of injured + killed	Float
injury_flag	1 if any injury occurred, else 0	Integer
fatality_flag	1 if any fatality occurred, else 0	Integer
severity	Fatal / Injury / Property Damage Only	Categorical

4.6. Analysis Dataset

A final analysis-ready subset was created by selecting only the analytically relevant columns and dropping rows where either **crash_date** or **crash_hour** was missing — as these temporal fields are foundational to most visualizations.

5. Exploratory Data Analysis (EDA) Using Seaborn

All visualizations were created with **Seaborn**. The following subsections describe each visualization, the chart type chosen, and its interpretation.

5.1 Collision Count by Borough

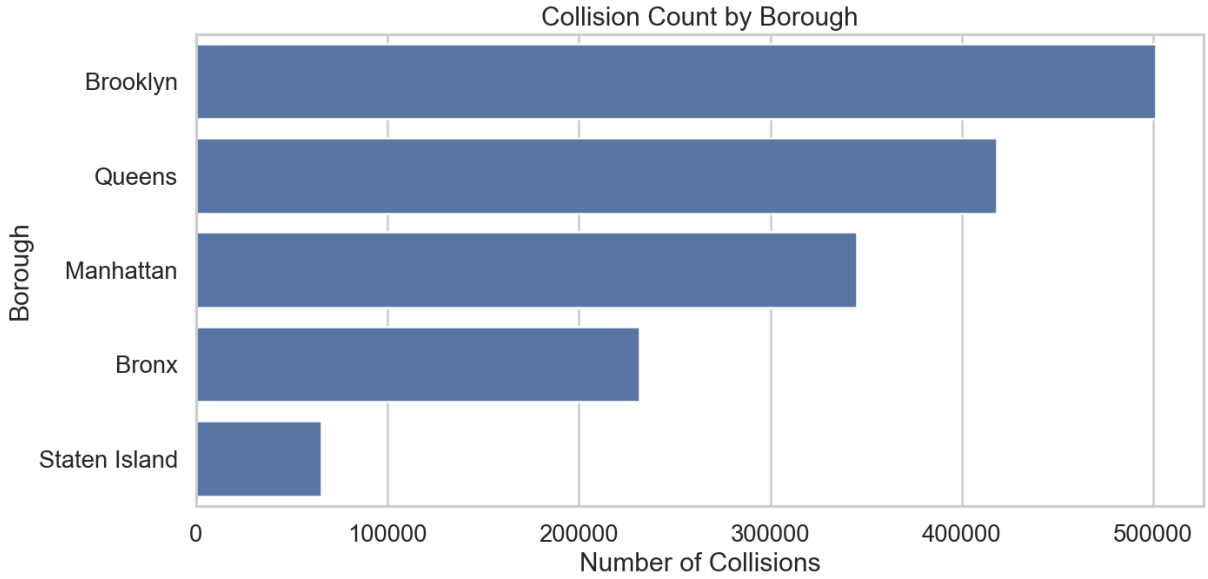

Figure 01 — Seaborn count plot (horizontal)

A horizontal count plot was used because the borough labels are long strings and the chart immediately reveals volume differences across the five NYC boroughs. The bars are ordered from highest to lowest collision count. High collision counts in specific boroughs may reflect a combination of population density, road network complexity, and traffic exposure rather than inherently unsafe driving behavior.

5.2 Distribution of Collisions by Hour of Day

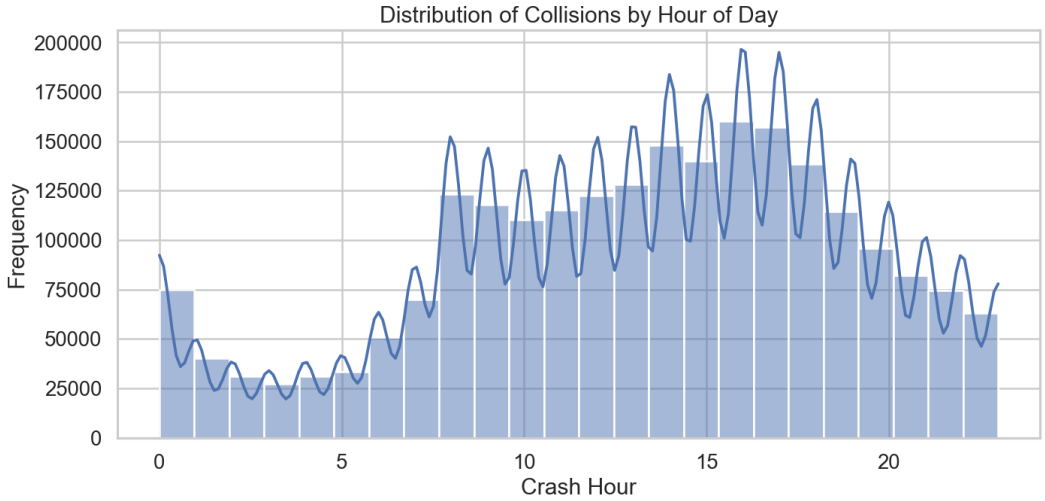

Figure 02 — Seaborn hist plot with KDE overlay

A histogram with 24 bins (one per hour) and a KDE overlay reveals the intraday collision rhythm. Two distinct peaks are typically visible: a morning commute peak (roughly 7–9 AM) and an afternoon/evening commute peak (roughly 4–6 PM). The KDE smooths the histogram to highlight the bimodal shape clearly.

5.3 Collision Count by Day of Week

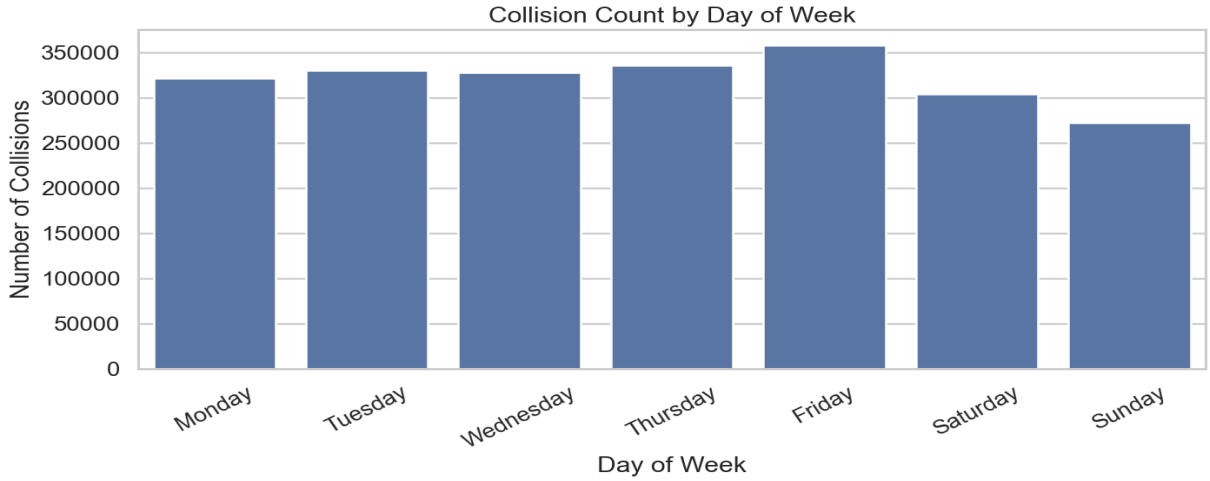

Figure 03 — Seaborn count plot (vertical, ordered Mon–Sun)

Days are ordered from Monday through Sunday so the viewer can immediately compare weekday versus weekend collision volumes. Weekdays tend to show higher absolute collision counts because of commuter traffic, while Saturdays may exhibit elevated late-night crash activity.

5.4 Collision Count by Month

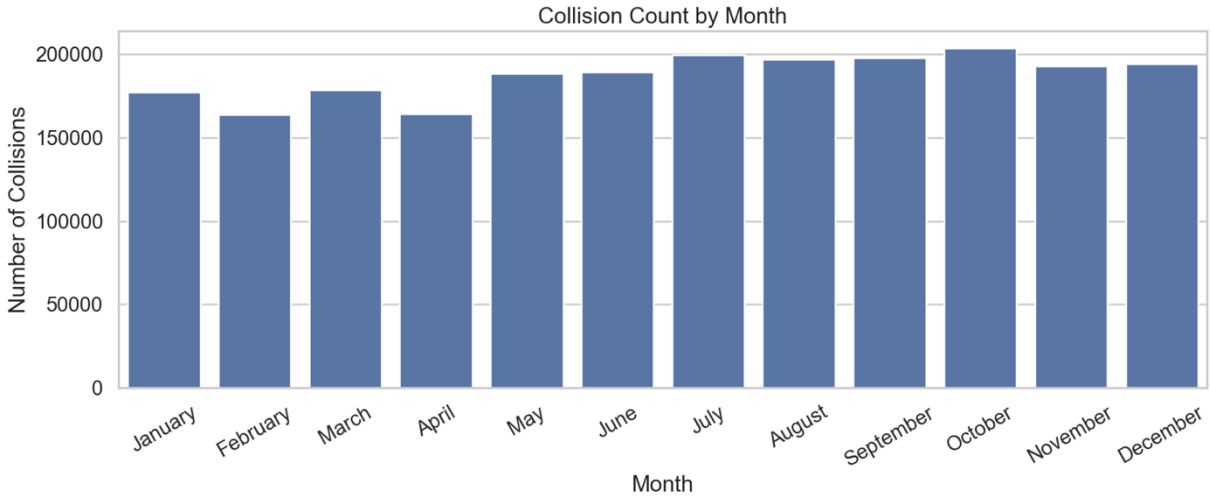

Figure 04 — Seaborn bar plot (ordered Jan–Dec)

A bar plot with months in calendar order reveals seasonal collision patterns across the full year. Warmer months (May–October) typically see higher collision volumes due to increased road usage. Winter months may show lower volumes but potentially higher severity due to adverse weather conditions.

5.5 Distribution of Persons Injured per Collision

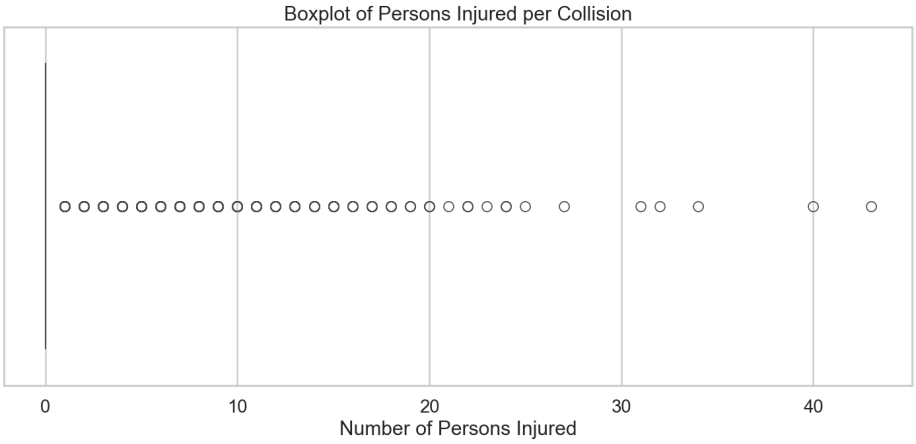

Figure 05 — Seaborn boxplot

The boxplot succinctly summarizes the injury distribution. Because most collisions result in zero or one injury, the median is expected to be near zero with a heavily right-skewed tail. Individual extreme outliers are visible as points beyond the whiskers, representing multi-vehicle pileups or other severe incidents.

5.6 Injury Distribution — Weekend vs Weekday

Figure 06 — Seaborn violin plot with quartile markers

The violin plot extends the boxplot by showing the full density shape of the injury distribution for weekday collisions (is_weekend=False) versus weekend collisions (is_weekend=True). Inner quartile lines reveal the concentration of values. Differences in shape between the two groups are examined further in the statistical testing section.

5.7 Collision Severity Distribution

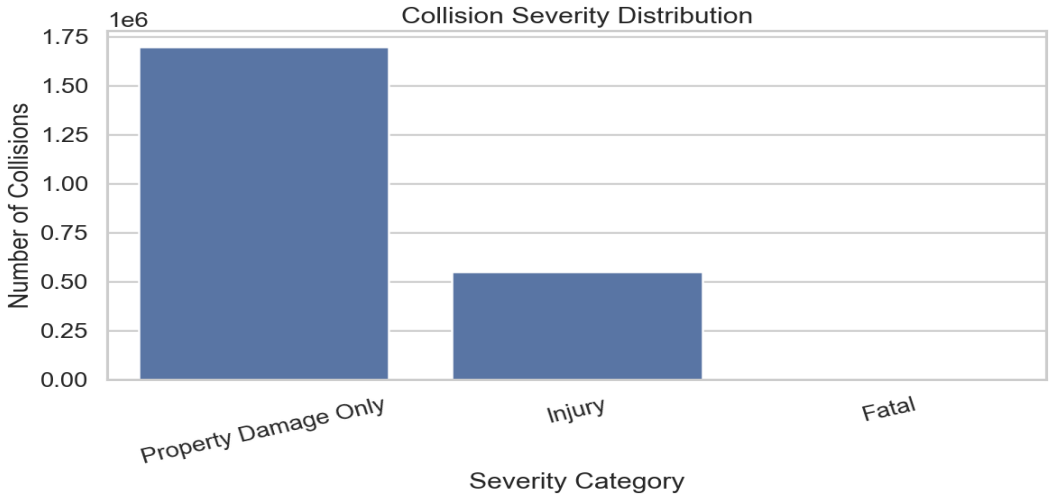

Figure 07 — Seaborn count plot (ordered by severity)

Collisions are classified into three mutually exclusive severity categories: Property Damage Only (no injuries), Injury, and Fatal. Most collisions fall into the Property Damage Only or Injury categories, with Fatal collisions representing a small but critically important minority.

5.8 Correlation Heatmap

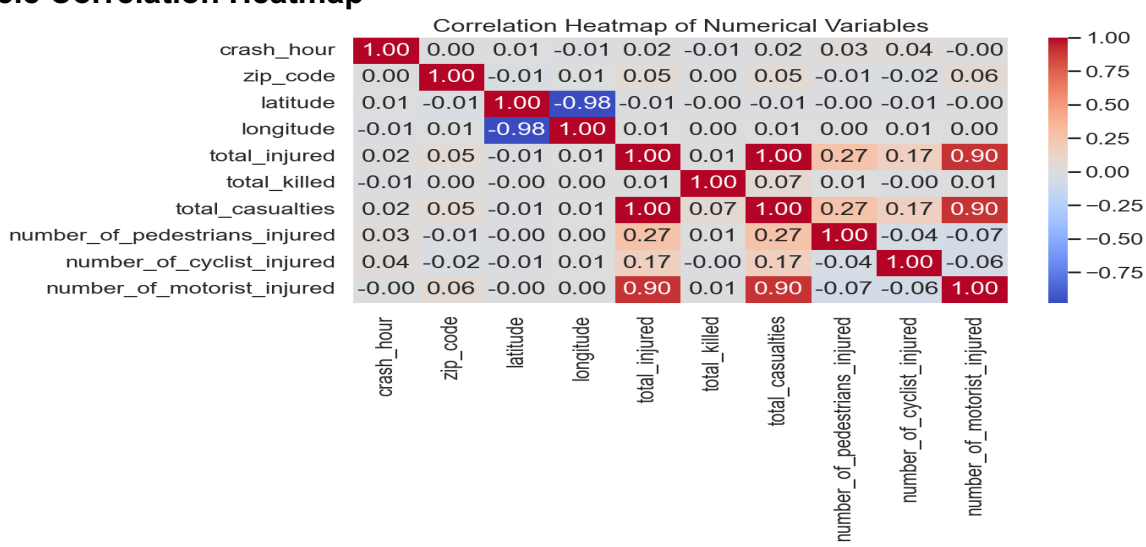

Figure 08 — Seaborn heatmap with annotated correlation coefficients

A Pearson correlation heatmap across all relevant numeric variables visualizes linear relationships briefly. Key observations include strong positive correlations between total_casualties and individual injury sub-categories (pedestrian, cyclist, motorist injured), which is expected given their additive relationship. Latitude and longitude show near-zero correlation with crash outcome variables, indicating that raw coordinates are weak predictors of severity on their own.

5.9 Top 10 Primary Contributing Factors

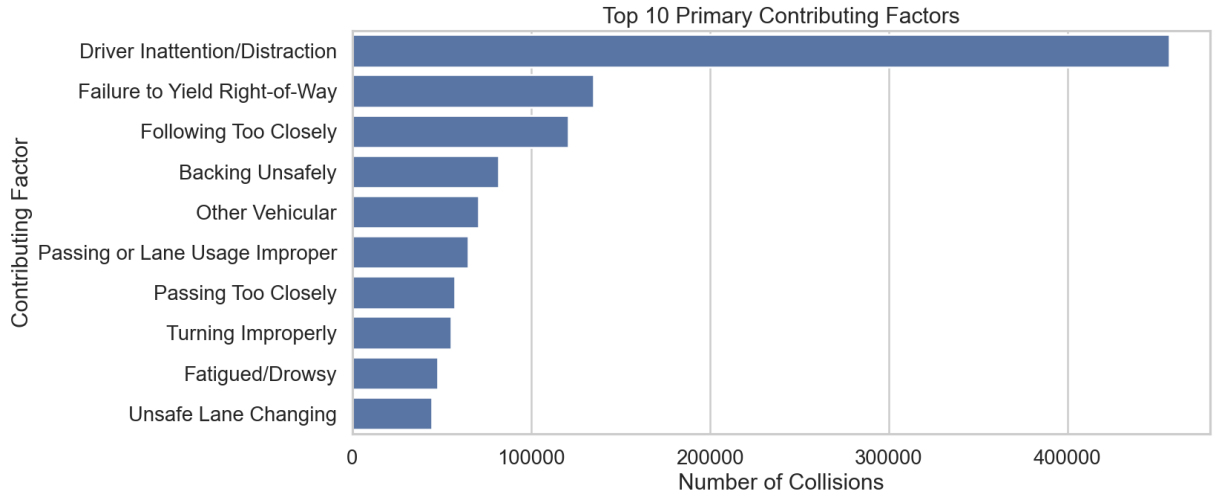

Figure 09 — Seaborn horizontal bar plot

The horizontal bar plot shows the ten most frequently cited contributing factors for the primary vehicle involved in each collision. Driver inattention/distraction and failure to yield are consistently among the top factors in NYC crash data, reflecting known national trends in distracted and inattentive driving.

5.10 Injuries by Borough (Boxplot)

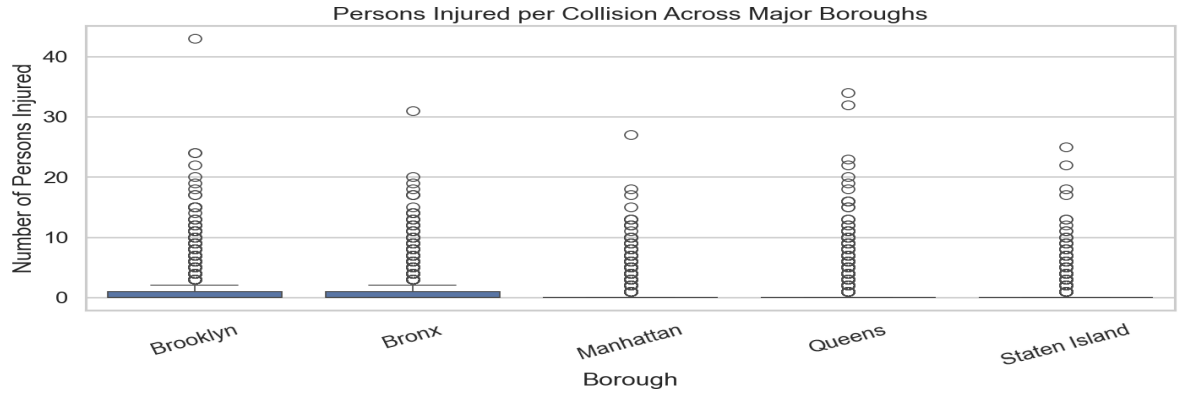

Figure 11 — Seaborn boxplot by borough

Comparing the injury distribution across the five major boroughs reveals whether per-collision severity differs geographically. While one borough may have more total collisions, another may have a higher median injury count per event — a distinction important for resource allocation in emergency response.

5.11 Collision Trend by Year

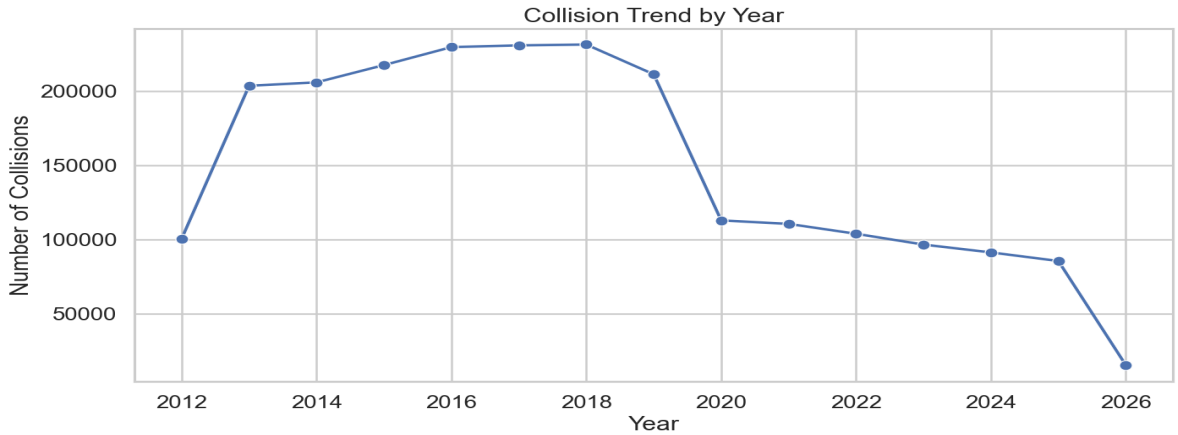

Figure 12 — Seaborn line plot with markers

The year-over-year trend line highlights long-term changes in collision frequency. A notable dip in 2020 is expected due to COVID-19 related reductions in traffic volume. Post-pandemic years may show a partial recovery toward pre-pandemic levels.

6. Statistical Analysis

Beyond visual inspection, two formal statistical tests were applied to evaluate hypotheses suggested by the EDA visualizations, along with descriptive statistics for key outcome variables.

6.1 Descriptive Statistics

Summary statistics (count, mean, standard deviation, min, quartiles, max) were computed for the following variables:

Variable	Description	Expected Pattern
total_injured	Persons injured per collision	Right-skewed; median near 0
total_killed	Persons killed per collision	Near-zero with rare large values
total_casualties	Injured + killed per collision	Right-skewed composite measure
crash_hour	Hour of day (0–23)	Bimodal with commute peaks

6.2 Mann–Whitney U Test — Weekday vs Weekend Injuries

A non-parametric Mann–Whitney U test was chosen over a t-test because the injury distribution is highly right-skewed and non-normal — a common characteristic of count-based safety outcomes. The test evaluates the null hypothesis that the distribution of injuries per collision is identical on weekdays and weekends.

Test	Mann–Whitney U (two-sided)

Groups	Weekday collisions vs Weekend collisions
out var.	total_injured (persons injured per collision)
Significance	$p < 0.05$ → reject null hypothesis
Interpretation	If significant: weekday and weekend injury distributions differ; if not: no detectable difference in

6.3 Chi-Square Test — Borough vs Collision Severity

A chi-square test of independence was applied to the contingency table of borough by severity category (Fatal, Injury, Property Damage Only). This tests whether the distribution of collision severity is homogeneous across boroughs, or whether certain boroughs have a disproportionately high share of fatal or injurious crashes.

Test	Chi-square test of independence
Row variable	Borough (5 categories)
Column var.	Severity (Fatal, Injury, Property Damage Only)
Significance	$p < 0.05$ → borough and severity are statistically associated
Interpretation	A significant result implies that collision severity patterns are not uniform ac boroug

7. Advanced Analysis (Optional)

Two additional analyses were performed beyond the required EDA to deepen the interpretation of collision patterns.

7.1 Average Casualties per Collision by Hour of Day

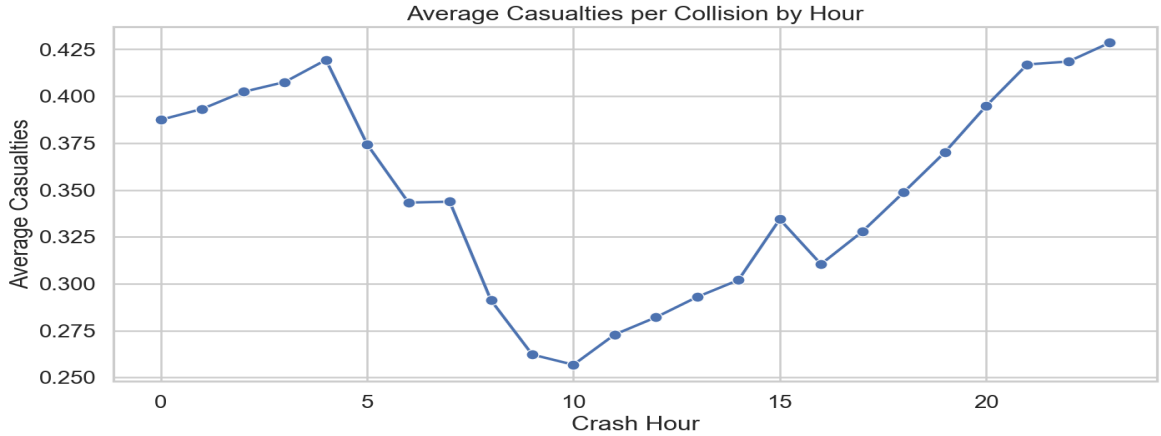

Figure 13 — Seaborn line plot with hourly aggregation

While the collision count histogram (Figure 02) shows when collisions are most frequent, this advanced chart shows when they are most severe. Average casualties per collision are computed for each of the 24 hours. Late-night hours (midnight–4 AM) often show elevated average casualties despite lower total collision counts — a pattern consistent with impaired driving, higher speeds on quieter roads, and reduced police visibility.

7.2 Fatal Collision Rate by Borough

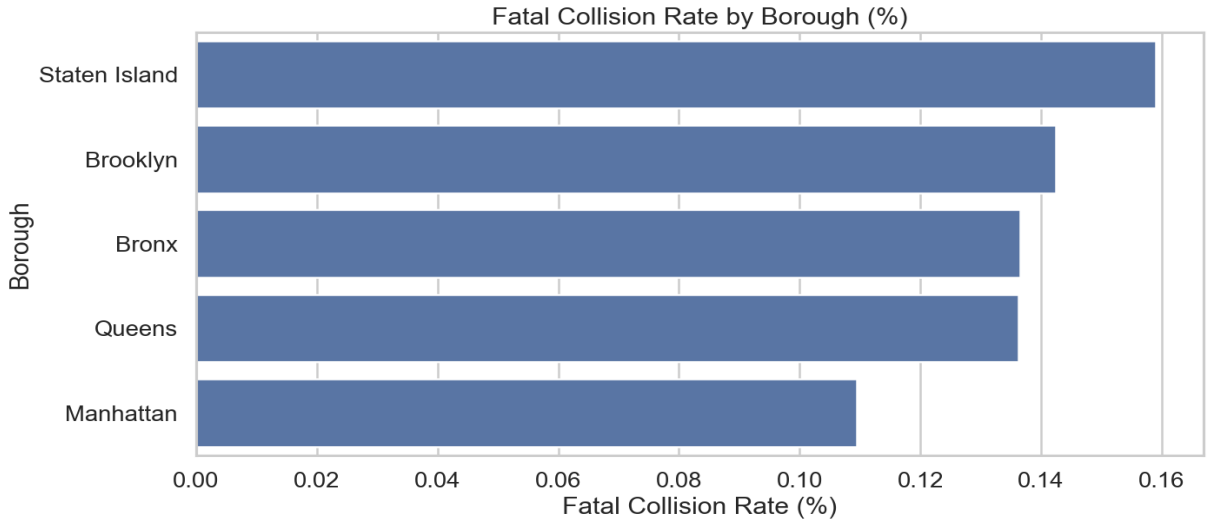

Figure 14 — Seaborn horizontal bar plot of borough-level fatality rates

The fatal collision rate is calculated as the number of collisions that resulted in at least one fatality divided by the total number of collisions in that borough, expressed as a percentage. This normalizes for the raw volume differences between boroughs revealed in Figure 01. A borough with fewer total collisions may still have a higher fatality rate, indicating more severe crash environments — potentially related to road design, speed limits, or the proportion of pedestrian-heavy areas.

8. Key Findings and Conclusions

The following evidence-based findings emerge from the combined EDA and statistical analysis of the NYC Motor Vehicle Collisions dataset:

Temporal Patterns

- **Bimodal daily distribution:** Collision frequency peaks during morning (7–9 AM) and evening (4–6 PM) commute hours, strongly suggesting that traffic exposure is the primary driver of collision frequency.
- **Weekday dominance:** Weekdays account for most collisions in absolute terms, but late-night weekend hours carry a disproportionately higher per-collision severity, likely linked to impaired or fatigued driving.

- **Seasonal effects:** Warmer months register higher collision volumes, consistent with increased road usage, more cyclists, and more pedestrian activity.
- **COVID-19 signature:** A pronounced decline in collision counts is visible in 2020, reflecting the dramatic reduction in urban mobility during the pandemic.

Geographic Patterns

- **Borough-level volume variation:** Some boroughs record substantially more collisions than others, which may reflect population density, road network density, and commuter throughput rather than inherently more dangerous driving.
- **Fatal collision rate disparities:** After normalizing for total collision count, the fatal collision rate varies across boroughs. Boroughs with more arterial roads, highway access, or lower pedestrian infrastructure investment may show elevated fatality rates.

Severity and Injury Patterns

- **Heavily skewed injury distribution:** Most collisions result in zero injuries (Property Damage Only), while a small number of events account for the bulk of total injuries — a characteristic right-skewed distribution confirmed by the boxplot.
- **Statistical significance of weekday/weekend difference:** The Mann–Whitney U test was applied to formally test whether injury distributions differ by day type. If the p-value was below 0.05, this confirms a statistically meaningful difference in injury severity between weekday and weekend collisions.
- **Borough–severity association:** The chi-square test confirmed that collision severity is not uniformly distributed across boroughs, indicating borough-specific risk profiles that could inform targeted interventions.

Contributing Factors

- **Driver inattention/distraction** is consistently the most frequently cited contributing factor, followed by failure to yield right-of-way and following too closely. These behavioral factors point to the potential impact of enforcement campaigns, public awareness programs, and in-vehicle safety technology.

Overall Conclusion

The NYC Motor Vehicle Collisions dataset is exceptionally well-suited for advanced Seaborn-based visualization and statistical analysis. Its combination of temporal granularity, spatial coverage, injury measurement, and causal attribution provides a comprehensive picture of urban traffic safety. The analysis demonstrates that collision patterns are strongly shaped by commuting behavior, geographic context, and behavioral factors — all of which represent actionable areas for public safety policy. Future work could integrate weather data, Vision Zero enforcement records, and speed camera locations to develop predictive models of high-risk collision events.